



TI-99/4A

TI BASIC (CONSOLE) · TI EXTENDED BASIC (CARTRIDGE) · 32×24 SCREEN

Conventions: [XB] marks Extended-BASIC-only features. **CALL** runs a built-in subprogram. **FCTN** = the gold function key. Colours are **1-16**; there is **no PEEK/POKE** in console BASIC.

PROGRAM CONTROL

NEW	Erase program
RUN	Start program
LIST	LIST / LIST "PIO" to printer
NUMBER (NUM)	Auto line numbers
RESEQUENCE (RES)	Renumber lines
EDIT n	Edit a line (or n & FCTN-↓)
OLD dev	Load: OLD DSK1.NAME
SAVE dev	SAVE DSK1.NAME
CONTINUE (CON)	Resume after BREAK
TRACE / UNTRACE	Show line numbers
BREAK / UNBREAK	Set / clear breakpoints
BYE	Exit to title screen
[XB] adds MERGE, RUN "DSK1.PROG", DELETE.	

STATEMENTS

[XB]=Ext. BASIC

PRINT	: new line, ; join, , zones
DISPLAY	[XB] DISPLAY AT(r,c):
INPUT	[XB] ACCEPT AT(r,c)
LET	Assign value
REM !	Comment (! is [XB] tail)
GOTO GOSUB	Jump / call; RETURN
ON x GOTO	/ GOSUB
FOR..NEXT	FOR I=1 TO 9 STEP 2
IF..THEN..ELSE	Branch
DATA READ	RESTORE [line] rewinds
DIM	Arrays; OPTION BASE 0/1
DEF f(x)=	Define function
RANDOMIZE	Seed RND
STOP / END	Halt

[XB] only: :: multi-statement, SUB/SUBEND, LINPUT, ON ERROR.

FUNCTIONS

ABS SGN INT	magnitude/sign/floor
SQR EXP LOG	root, e^x, ln
SIN COS TAN ATN	radians
RND	0-1 (RANDOMIZE seeds)
LEN POS	length / find substring
SEG\$(s\$,p,n)	substring (no LEFT\$/MID\$)
STR\$ VAL	num↔string
CHR\$ ASC	code↔char
TAB(n)	in PRINT

GRAPHICS CALLS

32×24 screen

CALL CLEAR	Clear screen to spaces
CALL SCREEN(c)	Background colour 1-16
CALL COLOR(s,f,b)	Char-set s fg/bg colour
CALL CHAR(n,p\$)	Define char from 16 hex
CALL HCHAR(r,c,n,x)	Place char across (x reps)
CALL VCHAR(r,c,n,x)	Place char down
CALL GCHAR(r,c,v)	Read char code at cell
Rows 1-24, cols 1-32. Char codes 32-95 ASCII; 96-127 small; 128-159 user.	

SPRITES & MOTION

[XB] only

CALL SPRITE(#n,c,col,r,cl[,rv,cv])	Create sprite #1-#28
CALL MOTION(#n,rv,cv)	Set row/col velocity
CALL POSITION(#n,r,c)	Read position
CALL LOCATE(#n,r,c)	Move sprite
CALL PATTERN(#n,ch)	Change shape
CALL MAGNIFY(n)	Size 1-4
CALL COINC / DISTANCE	Collision gap
CALL DELSPRITE(#n)	Remove (or ALL)

SOUND & INPUT CALLS

CALL SOUND(d,f,v,...)	dur ms (neg=cut), up to 3 tones+noise freq 110-44733 Hz · vol 0 loud-30 off
CALL KEY(u,k,s)	Read keypress: k=code, s=status
CALL JOYST(u,x,y)	Joystick: x,y = -4 / 0 / 4 Noise = negative freq (-1...-8). Unit 0 = whole keyboard for CALL KEY.

COLOUR SET

1-16

1 Transp.	2 Black
3 Med Grn	4 Lt Grn
5 Dk Blue	6 Lt Blue
7 Dk Red	8 Cyan
9 Med Red	10 Lt Red
11 Dk Yel	12 Lt Yel
13 Dk Grn	14 Magenta
15 Grey	16 White

Colour 1 (transparent) shows the screen colour behind. Chars are grouped 8 per colour set.

DEVICES & FILES

DSK1. DSK2.	Disk: DSK1.FILENAME
CS1 CS2	Cassette recorders
PIO	Parallel printer
RS232	Serial port
OPEN #1:"DSK1.D",..	type, mode, format
INPUT#1 / PRINT#1	Read / write records
RESTORE #1	Rewind file
CLOSE #1	Close file

File attr: SEQUENTIAL/RELATIVE, DISPLAY/INTERNAL, FIXED/VARIABLE, INPUT/OUTPUT/UPDATE/APPEND.

FCTN (GOLD) KEYS

FCTN + key

FCTN+E S D X	Cursor ↑ ← → ↓
FCTN+1 (DEL)	Delete character
FCTN+2 (INS)	Insert character
FCTN+3 (ERASE)	Erase to end of line
FCTN+4 (CLEAR)	Break / stop a RUN
FCTN+8 (REDO)	Recall last line
FCTN+9 (BACK)	Back / exit prompt
FCTN+= (QUIT)	To title — erases program!

Use **BYE**, not **QUIT**, to leave BASIC without losing your work.

EDITING TIPS

n TEXT	Add or replace line n
n <ENTER>	Delete line n
NUM	Turn on auto-numbering
RES	Renumber 100 step 10
EDIT n	Open line for editing

No **PEEK/POKE** in console TI BASIC — use [XB] **CALL PEEK/CALL LOAD** (needs 32K/Mini-Mem).

QUICK-FACTS

CPU	TMS9900 16-bit @ 3 MHz
Video	TMS9918A · 16K VRAM · 32×24
Sound	TMS9919 · 3 tones + noise
RAM	16K VDP + 256-byte CPU pad
BASIC	interpreted via GPL (slow)
Ext. BASIC	cartridge: faster + sprites